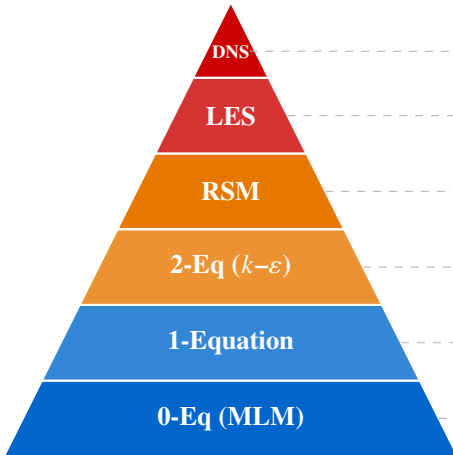


Empirical modelling / assumptions

Computational cost



Solves	Variables	Closure
Instantaneous N-S	u, v, w, p	None (exact)
Filtered N-S	$\bar{u}, \bar{v}, \bar{w}, \bar{p}$	Sub-grid-scale model
Mean flow + 7 PDEs	$U_i, P, \overline{u'_i u'_j}, \varepsilon$	Modelled stress transport
Mean flow + 2 PDEs	U_i, P, k, ε	Boussinesq EVM
Mean flow + 1 PDE	U_i, P, k	EVM + algebraic l_s
Mean-flow PDEs only	U_i, P	EVM + algebraic l_m